PROJECT PLANNING & MANAGEMENT

Store Sales Data Analysis

DEPI Graduation Project • Data Analysis Track

Dataset: Superstore Sales Dataset.csv

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Prepared for project planning and team review

1. Project Proposal

1.1 Project Overview

The project analyzes the Superstore Sales Dataset to understand sales performance and identify patterns across time, geographic regions, products, and customer segments. The work includes data preparation, analysis, KPI calculation, insight generation, and an interactive dashboard to communicate findings.

1.2 Problem Statement

Sales datasets contain information about orders, customers, products, and regions. Without structured analysis, it can be difficult to compare performance, identify trends, and discover useful patterns. This project turns the available records into clear findings that support data-informed decisions.

1.3 Project Objectives

- Analyze sales trends over time.
- Compare sales performance across geographic regions.
- Identify products and product categories with the highest sales.
- Analyze sales by customer segment.
- Calculate KPIs supported by the available data.
- Present findings through an interactive dashboard.

1.4 Project Scope

In scope: dataset inspection and quality assessment; data cleaning and preparation; analysis by date, region, product, and customer segment; KPI calculation; insight generation; interactive dashboard development; documentation; and GitHub file management.

Scope limitation: profit, profit margin, units sold, and discount amount will not be reported as actual measures unless reliable source fields or supplementary data become available.

2. Project Plan

2.1 Official Milestones

Phase	Official Deadline	Expected Deliverables
Project Planning & Management	16 Oct 2026	Planning documents
Literature Review	16 Oct 2026	Review and improvement notes
Requirements Gathering	16 Oct 2026	Requirements documentation
System Analysis & Design	6 Nov 2026	Analysis and design documents
Implementation (Source Code & Execution)	30 Nov 2026	Prepared data, analysis, dashboard
Final Presentation, Testing & Reports	4 Dec 2026	Testing records, reports, final presentation

2.2 Proposed Planning Schedule

Task	Proposed Dates	Output
Prepare Project Proposal	10–11 Oct	Proposal draft

Task	Proposed Dates	Output
Prepare Project Plan and Gantt schedule	11–12 Oct	Timeline and milestones
Define roles and responsibilities	11–12 Oct	Task assignment table
Prepare risk plan and KPIs	12–13 Oct	Risk and KPI sections
Review and submit planning documents	14–16 Oct	Reviewed planning package

The detailed dates above are proposed for internal organization; the official deadlines are listed separately in the milestones table.

2.3 Required Resources

Resource	Purpose
Superstore Sales Dataset.csv	Primary dataset
Excel / Power Query	Inspection, cleaning, and preparation
Power BI	Analysis and interactive dashboard
GitHub	Version control and file sharing
Team members and project documentation	Execution, review, and reporting

2.4 Project Monitoring

- Review task progress regularly.
- Track milestones and deliverables.
- Document contributions in GitHub.
- Report delays or blockers early.
- Review output quality before submission.

3. Task Assignment & Roles

Team size: 6 members. The role allocation below is a proposed structure and should be confirmed by the team. Replace Member 1–Member 6 with the actual names.

Phase	Team Members	Tasks & Responsibilities
Planning & Management	Mohamed Samir Mohamed Hassan (Team Leader) + Mahmoud Hassan	Project planning, timeline, task distribution, risk assessment, and KPIs.
Literature Review	Mohamed Hamdy + Ali Mamdouh	Reviewing project background, documenting findings, suggesting improvements, and preparing evaluation criteria.
Requirements Gathering	Ahmed Abd El Sattar + Habiba	Identifying stakeholders, user stories, use cases, and functional and non-functional requirements.
System Analysis & Design	Mohamed Samir Mohamed Hassan (Team Leader) + Mahmoud Hassan	Defining objectives, preparing ERD and DFD, reviewing use cases, designing wireframes, and selecting the technology stack.
Data Cleaning & Modeling	Mohamed Hamdy + Ali Mamdouh	Cleaning data, handling missing values and duplicates, validating data types, and building the data model and relationships.
DAX & Insights	Mohamed Samir Mohamed Hassan (Team Leader) + Mahmoud Hassan	Creating DAX measures, calculating KPIs, analyzing results, and extracting business insights and recommendations.
Dashboard & Visualization	Ahmed Abd El Sattar + Habiba	Designing dashboard pages, selecting charts, adding slicers and tooltips, and formatting visualizations.
Testing & Final Delivery	Mohamed Samir Mohamed Hassan (Team Leader) + All Team Members	Coordinating testing, checking results, fixing issues, reviewing documentation, and preparing the final presentation.

Collaboration and Accountability

- Each member is responsible for assigned tasks and documenting contributions.
- Members collaborate on dependent tasks and review one another’s work.
- Files and contributions are maintained in a shared GitHub repository.
- Each member should be able to explain their individual contribution during the final presentation.

4. Risk Assessment & Mitigation Plan

Risk	Potential Impact	Mitigation Strategy
Missing values	Incomplete information may affect analysis.	Inspect missing values and choose treatment based on the field and analytical need.
Incorrect data types	Date and numeric calculations may fail or become inaccurate.	Validate column types and test date conversions before analysis.
Postal-code formatting	Leading zeros may be lost if postal codes are treated as numbers.	Store postal codes as text when needed to preserve formatting.
Duplicate records	True duplicates may inflate counts or totals.	Investigate duplicates and confirm whether records are genuinely duplicated before removal.
Incorrect calculations	KPIs and visuals may be inaccurate.	Validate formulas and reconcile totals with the source data.
Missing profit or quantity fields	Some profitability and unit-based KPIs cannot be calculated directly.	Do not present these as actual measures without reliable source or supplementary data.
Task delays	Project milestones may be missed.	Track deadlines, assign owners, and report delays early.
Communication issues	Tasks may overlap or remain incomplete.	Clarify responsibilities and hold regular progress check-ins.
GitHub conflicts	Changes may be overwritten or conflict.	Use clear commit messages and review changes before merging.
Dashboard errors	Visuals may misrepresent results.	Test measures, filters, and totals before submission.

Risk Monitoring

Review data quality before analysis, record issues and corrective actions, validate metrics and visuals, and update the risk plan when new issues arise.

5. Key Performance Indicators (KPIs)

The KPIs below are proposed for the project. Some are headline metrics, while others are analytical breakdowns for charts and comparisons.

No.	KPI	Description	Calculation
1	Total Sales	Total recorded sales value.	SUM(Sales)
2	Sales Growth Rate (%)	Sales change compared with a previous period.	(Current Sales − Previous Sales) ÷ Previous Sales × 100
3	Total Orders	Number of unique orders.	DISTINCTCOUNT(Order ID)
4	Average Order Value (AOV)	Average sales value per unique order.	Total Sales ÷ Total Orders
5	Total Customers	Number of unique customers.	DISTINCTCOUNT(Customer ID)
6	Repeat Customer Rate (%)	Share of customers with more than one order in a defined period.	Repeat Customers ÷ Total Customers × 100
7	Average Orders per Customer	Average unique orders per customer.	Total Orders ÷ Total Customers
8	Sales per Customer	Average sales per unique customer.	Total Sales ÷ Total Customers
9	Average Daily Sales	Average sales per day in the selected period.	Total Sales ÷ Number of days in period
10	Sales by Region	Compare sales across regions.	SUM(Sales) by Region
11	Sales by Category	Compare sales across product categories.	SUM(Sales) by Category
12	Sales by Sub-Category	Compare sales across sub-categories.	SUM(Sales) by Sub-Category
13	Sales by Product	Compare sales by product.	SUM(Sales) by Product
14	Sales by Segment	Compare sales across customer segments.	SUM(Sales) by Segment
15	Sales by Ship Mode	Compare sales by shipping mode.	SUM(Sales) by Ship Mode

Implementation Notes

• Total Sales, Sales Growth Rate, Total Orders, Average Order Value, and Total Customers are candidates for headline KPI cards.
• Regional, product, sub-category, segment, and shipping-mode measures are generally better shown as charts or comparisons.
• Sales Growth Rate requires a defined comparison period and non-zero previous-period sales.
• Repeat Customer Rate requires a defined time window and a clear repeat-customer definition.
• Average Daily Sales must specify whether the denominator includes all calendar days or only selling days.

Data Limitations

The current dataset does not provide dedicated Profit, product-cost, Quantity, or Discount fields. Therefore, Total Profit, Profit Margin, Units Sold, and Discount Amount cannot be calculated reliably from this file alone. They should be added only if trustworthy supplementary data becomes available.

Final Submission Checklist

☐ Project Proposal
☐ Project Plan and schedule
☐ Task Assignment & Roles
☐ Risk Assessment & Mitigation Plan
☐ KPIs
☐ Review formatting and consistency
☐ Upload project documents to GitHub

Official Planning & Management deadline: 16 October 2026.